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Biosynthesis, function and metabolic engineering of plant volatile organic compounds

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Summary

Plants synthesize an amazing diversity of volatile organic compounds (VOCs) that facilitate interactions with their environment, from attracting pollinators and seed dispersers to protecting themselves from pathogens, parasites and herbivores. Recent progress in -omics technologies resulted in the isolation of genes encoding enzymes responsible for the biosynthesis of many volatiles and contributed to our understanding of regulatory mechanisms involved in VOC formation. In this review, we largely focus on the biosynthesis and regulation of plant volatiles, the involvement of floral volatiles in plant reproduction as well as their contribution to plant biodiversity and applications in agriculture via crop–pollinator interactions. In addition, metabolic engineering approaches for both the improvement of plant defense and pollinator attraction are discussed in light of methodological constraints and ecological complications that limit the transition of crops with modified volatile profiles from research laboratories to real-world implementation.

I. Biosynthesis of volatile plant secondary metabolites and its interconnection with primary metabolism

A relatively large group of plant natural products consists of volatile organic compounds (VOCs), lipophilic liquids with low molecular weight and high vapor pressure at ambient temperatures. Physical properties of these compounds allow them to freely cross cellular membranes and be released into the surrounding environment (Pichersky *et al.*, 2006). Over the years > 1700 VOCs have been identified from 90 different plant families belonging to both

angio- and gymnosperms (Knudsen *et al.*, 2006). Biosynthesis of VOCs depends on the availability of carbon, nitrogen and sulfur as well as energy provided by primary metabolism. Therefore, the availability of these building blocks has a major impact on the concentration of any secondary metabolite, including VOCs, demonstrating the high degree of connectivity between primary and secondary metabolism. Biosynthesis of the wide array of different VOCs branches off from only a few primary metabolic pathways (Fig. 1). Based on their biosynthetic origin, all VOCs are divided into several classes, including terpenoids,

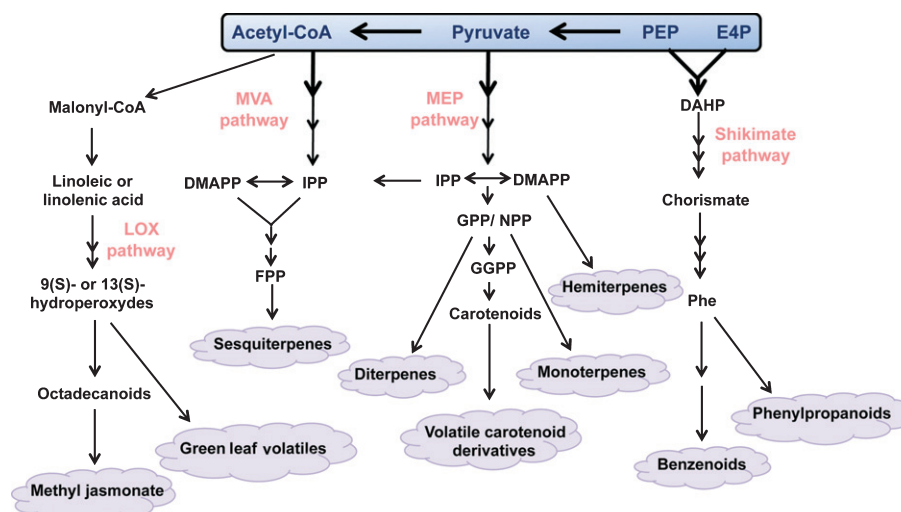


Fig. 1 Overview of biosynthetic pathways leading to the emission of plant volatile organic compounds (VOCs). Precursors for plant VOCs originate from primary metabolism (represented in the blue box). The four major VOC biosynthetic pathways, namely the shikimate/phenylalanine, the mevalonic acid (MVA), the methylerythritol phosphate (MEP) and lipoxygenase (LOX) pathways lead to the emission of benzenoids/phenylpropanoids, sesquiterpenes, monoterpenes, hemiterpenes, diterpenes, volatile carotenoid derivatives and methyl jasmonate/green leaf volatiles. Stacked arrows illustrate the involvement of multiple enzymatic reactions. VOCs are highlighted with a purple cloud as background. Abbreviations: DAHP, 3-deoxy-D-arabinoheptulosonate-7 phosphate; DMAPP, dimethylallyl pyrophosphate; E4P, erythrose 4-phosphate; FPP, farnesyl pyrophosphate; GGPP, geranylgeranyl pyrophosphate; GPP, geranyl pyrophosphate; IPP, isopentenyl pyrophosphate; NPP, neryl pyrophosphate; PEP, phosphoenolpyruvate; Phe, phenylalanine.

phenylpropanoids/benzenoids, fatty acid derivatives and amino acid derivatives in addition to a few species-/genus-specific compounds not represented in those major classes.

1. Biosynthesis of terpenoids

Terpenoids constitute the largest and most diverse class of secondary metabolites with many volatile constituents, which are derived from two common five-carbon precursors, isopentenyl diphosphate (IPP) and its allylic isomer, dimethylallyl diphosphate (DMAPP) (McGarvey & Croteau, 1995). In plants, two independent, compartmentally separated pathways – the mevalonic acid (MVA) and methylerythritol phosphate (MEP) – are responsible for the formation of these C₅-isoprene building units (Fig. 2). The MVA pathway gives rise to volatile sesquiterpenes (C₁₅), while the MEP pathway provides precursors to volatile hemiterpenes (C₅), monoterpenes (C₁₀) and diterpenes (C₂₀). The MEP pathway is considered to be exclusively plastidic, as a full set of the corresponding enzymes exists only in plastids based on experimental evidence and predictions of their subcellular localization (Hsieh *et al.*, 2008). By contrast, subcellular localization of the MVA pathway is not as clear. Historically, this pathway was referred to as being cytosolic; however, new evidence suggests that the MVA pathway is distributed between the cytosol, endoplasmic reticulum and peroxisomes (Simkin *et al.*, 2011; Pulido *et al.*, 2012).

The MVA pathway consists of six enzymatic reactions and is initiated by a stepwise condensation of three molecules of acetyl-CoA to 3-hydroxy-3-methylglutaryl-CoA, which undergoes reduction to MVA followed by two subsequent phosphorylations and a decarboxylation/elimination step with formation of IPP as the final product (Fig. 2) (Lange *et al.*, 2000). To date, it is still unclear which subcellular pool of acetyl-CoA is used for terpeneoid

biosynthesis, as acetyl-CoA cannot readily cross membranes, and pools exist in chloroplasts, peroxisomes, mitochondria, cytosol and nucleus (Oliver *et al.*, 2009). The Arabidopsis genome contains two genes encoding acetoacetyl-CoA thiolase (AACT), one of which (AACT2) catalyzes the first step in the MVA pathway (Ahumada *et al.*, 2008) and, based on proteome analysis, is localized in peroxisomes (Reumann *et al.*, 2007).

The MEP pathway involves seven enzymatic steps and begins with the condensation of D-glyceraldehyde 3-phosphate (GAP) and pyruvate (Pyr) to produce 1-deoxy-D-xylulose 5-phosphate, which is then subjected to isomerization/reduction with formation of the pathway's characteristic intermediate, MEP (Fig. 2). Five consecutive steps are required to convert MEP to IPP and DMAPP. The MEP pathway relies on primary metabolism for the supply of Pyr and GAP, with the latter derived from both glycolysis and the pentose phosphate pathway (PPP). To date, the origin of Pyr in the chloroplasts is not fully understood, as plastids have low activities of the key glycolytic enzymes, phosphoglycerate mutase and enolase (Andriotis *et al.*, 2010; Joyard *et al.*, 2010; Bayer *et al.*, 2011), and might not be able to sustain high Pyr demand for isoprenoid biosynthesis. Indeed, plastid-localized IPP biosynthesis was affected in *Arabidopsis thaliana* mutants lacking a Pyr transporter, which supplies cytosolic Pyr to the MEP pathway (Furumoto *et al.*, 2011).

Both IPP and DMAPP are substrates for short-chain prenyltransferases, which produce prenyl diphosphate precursors, geranyl diphosphate (GPP), farnesyl diphosphate (FPP) and geranylgeranyl diphosphate (GGPP), for a large family of terpene synthases/cyclases (TPSs) (Fig. 2) (Cane, 1999; Wise & Croteau, 1999). While the MVA pathway produces only IPP, the MEP pathway results in the synthesis of both IPP and DMAPP at a 6 : 1 ratio (Rohdich *et al.*, 2003). Thus, both pathways rely on

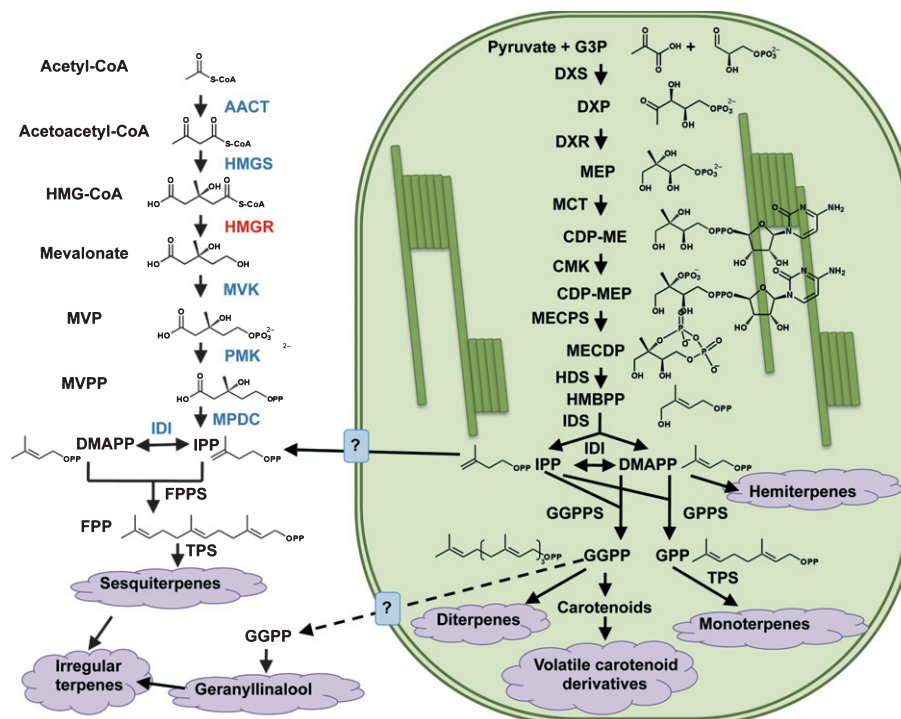


Fig. 2 Synthesis of terpenoid volatile organic compounds (VOCs). Terpenoid VOCs are synthesized by the cytosolic mevalonic acid (MVA) and the plastidial methylerythritol phosphate (MEP) pathways, the former giving rise to sesquiterpenes, irregular terpenes and geranylinalool, and the latter to monoterpenes, hemiterpenes, diterpenes and volatile carotenoid derivatives. Cross-talk between both pathways is facilitated by the export of isopentenyl pyrophosphate (IPP) from the plastid to the cytosol. Solid arrows represent established biosynthetic steps, whereas broken arrows point to hypothetical reactions. Volatile terpenoid compounds are highlighted with a purple cloud as background. Enzymes shown in blue exhibit peroxisomal localization, whereas the one in red is localized at the endoplasmic reticulum (ER). Abbreviations: AACT, acetyl-CoA acetyltransferase; CDP-ME, 4-diphosphocytidyl-2-C-methyl-D-erythritol; CDP-MEP, CDP-ME 2-phosphate; CMK, 4-(cytidine 5'-diphospho)-2-C-methyl-D-erythritol kinase; DMAPP, dimethylallyl pyrophosphate; DXP, 1-deoxy-D-xylulose 5-phosphate; DXS, DXP synthase; DXR, 1-deoxy-D-xylulose 5-phosphate reductoisomerase; FPP, farnesyl pyrophosphate; FPPS, FPP synthase; G3P, glyceraldehyde 3-phosphate; GGPP, geranylgeranyl pyrophosphate; GGPPS, GGPP synthase; GPP, geranyl pyrophosphate; GPPS, GPP synthase; HDS, 4-hydroxy-3-methylbut-2-en-1-yl diphosphate synthase; HMBPP, (E)-4-hydroxy-3-methylbut-2-en-1-yl diphosphate; HMG-CoA, hydroxymethylglutaryl-CoA; HMGR, HMG-CoA reductase; HMGS, HMG-CoA synthase; IDI, isopentenyl pyrophosphate isomerase; IDS, isopentenyl diphosphate synthase; IPP, isopentenyl pyrophosphate; MCT, 2-C-methyl-D-erythritol 4-phosphate cytidyltransferase; MECDP, 2-C-methyl-D-erythritol 2,4-cyclodiphosphate; MECPS, MECDP synthase; MVK, mevalonate kinase; MPDC, mevalonate diphosphate decarboxylase; MVP, mevalonate 5-phosphate; MVPP, mevalonate 5-pyrophosphate; PMK, phosphomevalonate kinase; TPS, terpene synthase.

isopentenyl diphosphate isomerase (IDI), which reversibly converts IPP to DMAPP (Nakamura *et al.*, 2001) and controls the equilibrium between them. Recently, in *Catharanthus roseus* and *Arabidopsis thaliana* genomes, a single and two distinct IDI genes were identified, respectively, which are transcribed as splice variants. In both plants, the 'long' proteins are transported into mitochondria and/or chloroplasts whereas the 'short' proteins, lacking the targeting signal, are located in the peroxisomes (Guirmand *et al.*, 2012), highlighting the involvement of different subcellular compartments in isoprenoid biosynthesis in plants.

IPP, DMAPP, and short prenyl diphosphates (GPP and FPP) facilitate the metabolic crosstalk between the compartmentally separated MVA and MEP pathways by acting as connecting metabolites (Nabeta *et al.*, 1997; Adam *et al.*, 1999; Hemmerlin *et al.*, 2003; Wu *et al.*, 2006; Orlova *et al.*, 2009). Trafficking of these compounds across the inner envelope membrane of plastids is mediated by an unidentified metabolite transporter (Bick & Lange, 2003; Flügge & Gao, 2005). Such connectivity of the isoprenoid biosynthetic pathways allows the MEP pathway, often with a higher

carbon flux than the MVA route, to support biosynthesis of terpenoids in the cytosol (Laule *et al.*, 2003; Dudareva *et al.*, 2005; Ward *et al.*, 2011). The contribution of these pathways to terpenoid biosynthesis is species- and/or organ-specific. While in snapdragon flowers, for example, the MEP pathway provides precursors for cytosolic sesquiterpene formation (Dudareva *et al.*, 2005), in carrot leaves and roots, sesquiterpenes are derived from both MEP and MVA pathways (Hampel *et al.*, 2005).

Despite the presence of IPP and DMAPP in several compartments, biosynthesis of prenyl diphosphate intermediates is compartment-specific and depends on subcellular localization of corresponding short-chain prenyltransferases. In the cytosol, the sequential head-to-tail condensation of two IPP molecules with one molecule of DMAPP catalyzed by FPP synthase gives rise to FPP, the precursor of volatile sesquiterpenes. In plastids GPP and GGPP synthases are responsible for the head-to-tail condensation of one DMAPP molecule with one or three IPP molecules to form GPP and GGPP, respectively, the corresponding precursors of mono- and diterpenes.

The tremendous diversity of volatile terpenoids in plants is generated through the action of TPSs (Fig. 2), many of which have the distinctive ability to synthesize multiple products from a single prenyl diphosphate substrate (Degenhardt *et al.*, 2009b). Indeed, in *Arabidopsis* two sesquiterpene synthases, TPS21 and TPS11, account for the biosynthesis of nearly all 20 sesquiterpenes found in the floral volatile blend (Tholl *et al.*, 2005). In addition, many TPSs accept more than one substrate (Tholl, 2006; Bleeker *et al.*, 2011), which expands the diversity of produced terpenoids by directing bifunctional enzymes to different compartments with a varying range of available substrates (Aharoni *et al.*, 2004; Nagegowda *et al.*, 2008; Huang *et al.*, 2012; Gutensohn *et al.*, 2013). Moreover, it has recently been shown that tomato monoterpene and sesquiterpene synthases can use prenyl diphosphates with a *cis* configuration, such as neryl diphosphate (NDP) and *Z,Z*-FPP, respectively, instead of the usual GPP and *E,E*-FPP, to form new products (Sallaud *et al.*, 2009; Schillmiller *et al.*, 2009; Bleeker *et al.*, 2011). However, it still remains to be determined if this is a general property of TPSs.

To date, the TPS gene family consists of >100 members characterized from many plant species, with about one-third isolated from flowers and fruits. This gene family has been divided into seven subfamilies (designated TPS-a through TPS-g) based on sequence relatedness, functional assessment, and gene architecture (Bohlmann *et al.*, 1998; Aubourg *et al.*, 2002). Interestingly, TPSs from related plant species tend to cluster together more than enzymes of similar function, thus challenging substrate/product predictions based on sequence similarities (Bohlmann *et al.*, 1998).

In addition to a wide range of volatile terpenoids formed directly by TPSs, terpenoid diversity is further increased by other enzymes that are capable of modifying the TPS products via hydroxylation, dehydrogenation, acylation, or other reactions, thus increasing their volatility and altering their olfactory properties (Dudareva *et al.*, 2004). Plants also produce irregular volatile terpenoids with carbon skeletons ranging from C₈ to C₁₈, which originate from carotenoids via three step modifications, including an initial dioxygenase cleavage followed by enzymatic transformation and acid-catalyzed conversion to volatile compounds (Fig. 2) (Winterhalter & Rouseff, 2001). In some cases, including *Arabidopsis*, tomato, petunia and melon, the dioxygenase cleavage step itself can yield a volatile product, such as α - and β -ionone, geranylacetone, and pseudoionone, from an array of carotenoid pigments (Simkin *et al.*, 2004; Ibdah *et al.*, 2006). The irregular acyclic C₁₁- and C₁₆-homoterpenes, 4,8-dimethylnona-1,3,7-triene (DMNT) and 4,8,12-trimethyltrideca-1,3,7,11-tetraene (TMTT), are derived from FPP and GGPP, respectively (Fig. 2). Their biosynthesis proceeds via two enzymatic steps, the formation of tertiary C₁₅- and C₂₀- alcohol precursors, (*E*)-nerolidol and (*E,E*)-geranyl linalool, respectively, followed by an oxidative degradation catalyzed by a cytochrome P450 monooxygenase (Tholl *et al.*, 2011).

2. Biosynthesis of phenylpropanoid/benzenoid compounds

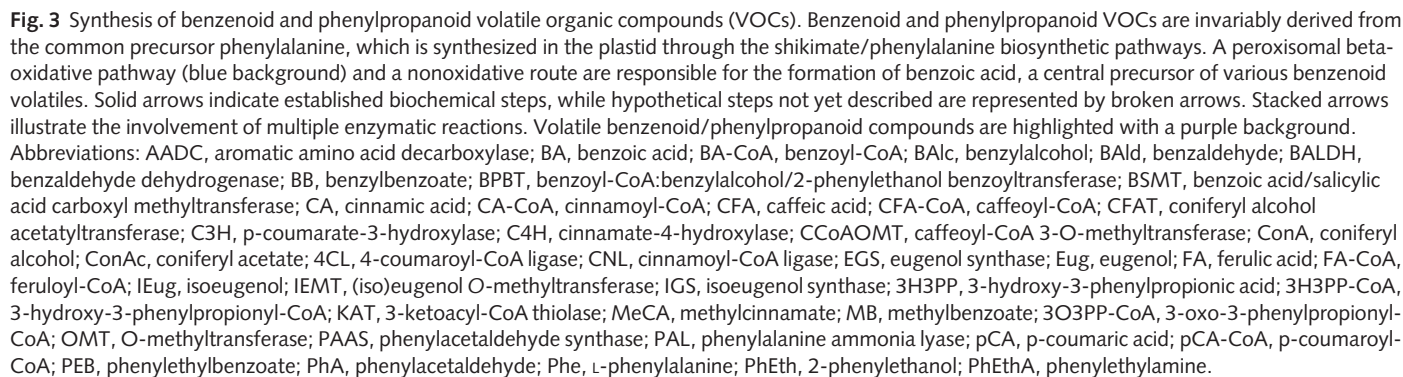
The second largest class of plant VOCs comprises phenylpropanoid and benzenoid compounds (Knudsen *et al.*, 2006), which originate from the aromatic amino acid phenylalanine (Phe)

(Fig. 3). Seven enzymatic reactions of the shikimate pathway and three of the arogenate pathway connect central carbon metabolism to Phe (Tzin & Galili, 2010; Maeda & Dudareva, 2012). The immediate precursors of the shikimate pathway, phosphoenolpyruvate (PEP) and D-erythrose 4-phosphate (E4P), derive from glycolysis and the PPP, respectively. The same pathways provide precursors for the MEP pathway, and thus the latter has to compete for carbon allocation with the shikimate/phenylpropanoid pathway, especially as 30% of photosynthetically fixed carbon is directed to Phe, largely to make lignin (Razal *et al.*, 1996). The first gene in the shikimate pathway, 3-deoxy-D-arabino-heptulosonate 7-phosphate synthase (DAHPS synthase), plays a key role in controlling carbon flux into the pathway (Tzin *et al.*, 2012). However, the molecular mechanisms involved in this regulation remain largely unknown in plants (Maeda & Dudareva, 2012).

While Phe biosynthesis takes place in plastids (Maeda & Dudareva, 2012), its further conversion to volatile compounds occurs outside this organelle. The first committed step in the biosynthesis of the majority of phenylpropanoids/benzenoids is catalyzed by a well-known and widely distributed enzyme, L-phenylalanine ammonia lyase (PAL), which deaminates Phe to *trans*-cinnamic acid (CA) (Fig. 3). Formation of the benzenoids (C₆-C₁) from cinnamic acid requires shortening of the propyl side chain by two carbons and has been shown to proceed via a β -oxidative pathway, a non β -oxidative pathway, or a combination of these pathways (Fig. 3) (Boatright *et al.*, 2004; Orlova *et al.*, 2006). The β -oxidative route was only recently fully elucidated and appears to be analogous to that operating in catabolism of fatty acids and certain branched-chain amino acids. This pathway begins with activation of CA to its CoA thioester, which undergoes hydration, oxidation and cleavage of the β -keto thioester, resulting in the formation of benzoyl-CoA (Van Moerkercke *et al.*, 2009; Klempien *et al.*, 2012; Qualley *et al.*, 2012). The peroxisomal localization of the β -oxidative pathway raises the question about the export of benzoyl-CoA to the cytosol for benzylbenzoate and phenylethylbenzoate biosynthesis.

The alternative non β -oxidative pathway proceeds via benzaldehyde as a key intermediate followed by its oxidation to benzoic acid (Fig. 3). While NAD⁺-dependent benzaldehyde dehydrogenase converting benzaldehyde to benzoic acid was isolated and characterized (Long *et al.*, 2009), biochemical steps leading to benzaldehyde formation are still in question. During the last decade, significant progress has been made in the discovery of enzymes and genes involved in the final steps of benzenoid volatile formation. Two enzyme superfamilies, BAHD superfamily of acyltransferases (D'Auria, 2006) and the SABATH family of methyltransferases (D'Auria *et al.*, 2003), were found to greatly contribute to the final biosynthetic steps of volatile benzenoids.

The volatile phenylpropenes (C₆-C₃), such as eugenol, isoeugenol, methyleugenol, isomethyleugenol, chavicol, and methylchavicol, share the initial biosynthetic steps with the lignin biochemical pathway up to the phenylpropenol (monolignol) stage and then require two enzymatic reactions to eliminate the oxygen functionality at C-9 position (Koeduka *et al.*, 2006; Dexter *et al.*, 2007). Coniferyl alcohol is first converted to coniferyl acetate by coniferyl alcohol acetyltransferase (Dexter *et al.*, 2007) before its reduction



In contrast to benzenoids and phenylpropenes, the biosynthesis of volatile phenylpropanoid-related (C₆-C₂) compounds, such as phenylacetaldehyde and 2-phenylethanol, does not occur via CA and competes with PAL for Phe utilization (Fig. 3) (Kaminaga *et al.*, 2006; Tieman *et al.*, 2007). In petunia and rose petals, phenylacetaldehyde is produced directly from Phe via an unusual combined decarboxylation-amine oxidation reaction catalyzed by

Another class of plant VOCs comprises fatty acid derivatives such as 1-hexanal, *cis*-3-hexenol, nonanal and methyl jasmonate, which

arise from C₁₈ unsaturated fatty acids, linoleic or linolenic (Fig. 4). Biosynthesis of these fatty acids relies on a plastidic pool of acetyl-CoA generated from Pyr, the final product of glycolysis. After entering the 'lipoxygenase (LOX) pathway', unsaturated fatty acids undergo stereospecific oxygenation to form 9-hydroperoxy and 13-hydroperoxy intermediates (Fig. 4) (Feussner & Wasternack, 2002), which are further metabolized via the two branches of the LOX pathway yielding volatile compounds. The allene oxide synthase branch uses only the 13-hydroperoxy intermediate and leads to the formation of jasmonic acid (JA), which in turn is converted to methyl jasmonate by JA carboxyl methyl transferase (Song *et al.*, 2005). By contrast, the hydroperoxide lyase branch converts both types of hydroperoxide fatty acid derivatives into C₆ and C₉ aldehydes, which are often reduced to alcohols by alcohol dehydrogenases (Gigot *et al.*, 2010), followed by further conversion to their esters (Fig. 4) (D'Auria *et al.*, 2007). These saturated and unsaturated C₆/C₉ aldehydes and alcohols commonly referred to as green leaf volatiles, are usually synthesized in green organs of plants in response to wounding, but also provide fruits and vegetables with their characteristic 'fresh green' aroma.

4. Biosynthesis of volatiles derived from branched-chain amino acids

Numerous volatile compounds, especially those highly abundant in floral scents and fruit aromas, are derived from amino acids such as alanine, valine, leucine, isoleucine, and methionine, or intermediates in their biosynthesis, and contain nitrogen and sulfur (Fig. 5) (Knudsen *et al.*, 2006). The biosynthesis of these amino acid-derived volatiles in plants is believed to proceed in a similar way to that found in bacteria or yeast (Dickinson *et al.*, 2000; Beck *et al.*, 2002; Tavaría *et al.*, 2002), where these pathways have been studied more extensively. As in microorganisms, the amino acids undergo an initial deamination or transamination catalyzed by aminotransferases, leading to the formation of the corresponding α -ketoacid (Fig. 5) (Gonda *et al.*, 2010). These α -ketoacids can be further subjected to decarboxylation, followed by reductions, oxidations and/or esterifications, forming aldehydes, acids, alcohols and esters (Fig. 5) (Reineccius, 2006). Amino acids can also be the precursors of acyl-CoAs, which are used in alcohol esterification reactions catalyzed by alcohol acyltransferases (AATs) (Beekwilder *et al.*, 2004; Gonzalez *et al.*, 2009).

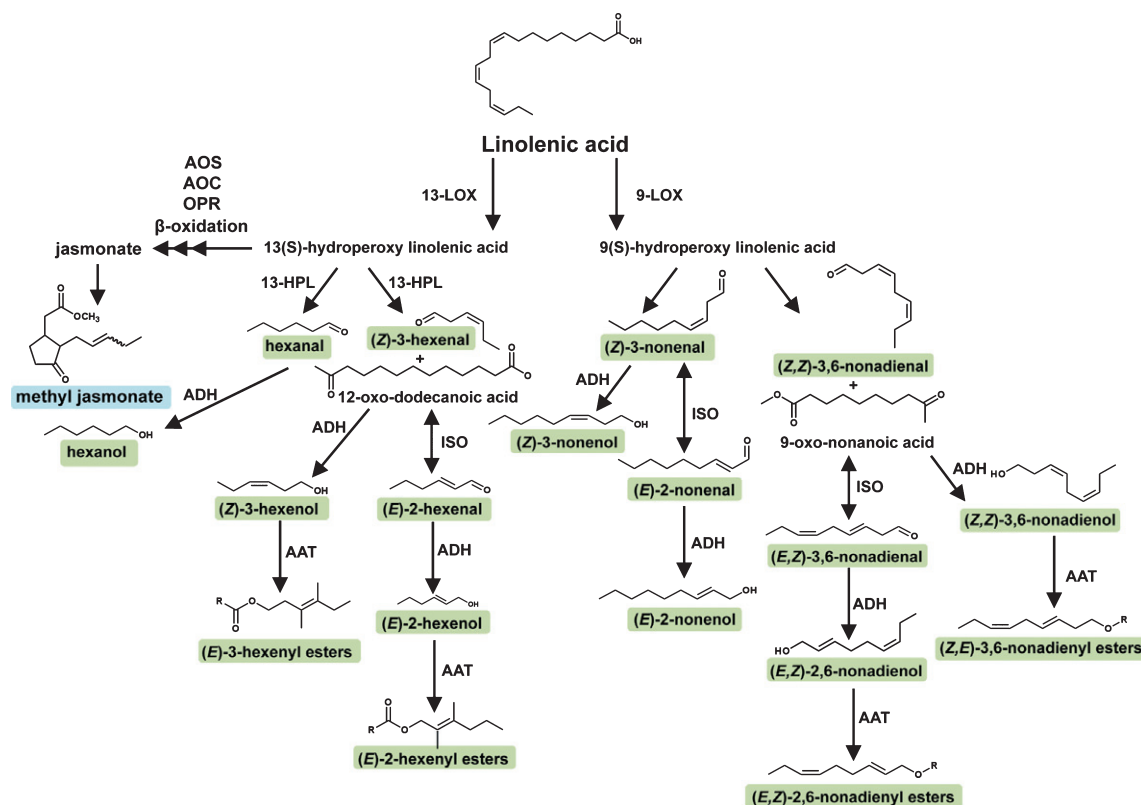


Fig. 4 Synthesis of fatty acid-derived volatile organic compounds (VOCs) and green leaf volatiles (GLVs). Linoleic and linolenic acid serve as the precursors for a variety of fatty acid-derived VOCs. These precursors enter the lipoxygenase (LOX) pathway by an oxidation yielding 9-hydroperoxy and 13-hydroperoxy intermediates that are further converted to volatiles by hydroperoxide lyases and alcohol dehydrogenases. Stacked arrows illustrate the involvement of multiple enzymatic reactions. Volatile compounds are highlighted with a colored background, with green leaf volatiles shown in green. Abbreviations: AAT, alcohol acyltransferase; ADH, alcohol dehydrogenase; AOC, allene oxide cyclase; AOS, allene oxide synthase; 9-HPL, 9-hydroperoxide lyase; 13-HPL, 13-hydroperoxide lyase; ISO, isomerase; 9-LOX, 9-lipoxygenase; 13-LOX, 13-lipoxygenase; OPR, 12-oxophytodienoate reductase.

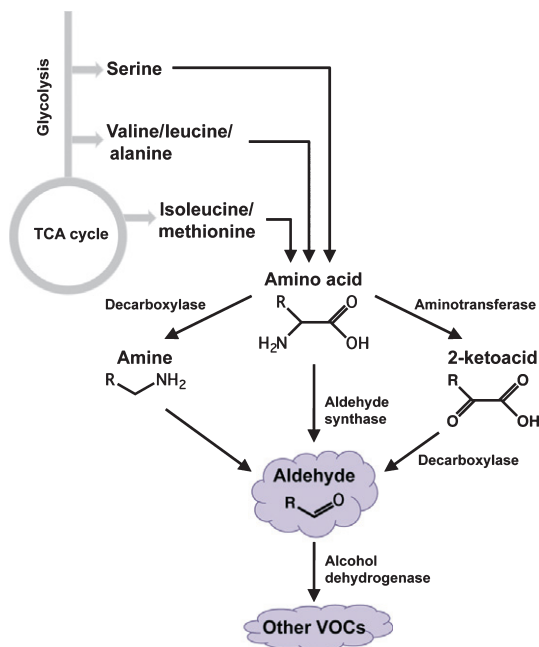


Fig. 5 Synthesis of volatile organic compounds (VOCs) derived from branched-chain amino acids. The formation of VOCs from branched-chain amino acids proceeds through different parallel routes. The aldehyde can be synthesized by decarboxylation of the amino acid followed by deamination; by direct synthesis of the aldehyde; and by transamination of the amino acid followed by a carboxylation. The aldehyde is further converted to a variety of VOCs with diverse functional groups. TCA, tricarboxylic acid.

II. Regulation of volatile emission in plants

In plants, formation of VOCs is spatially regulated. While the most diverse mixture with the highest abundance of VOCs is released predominantly from flowers in the majority of flowering plants (Knudsen *et al.*, 2006), vegetative tissues, such as foliage and stems in conifers, and the leaves of herbs, such as peppermint and basil, represent the main source of VOC emission in these species (McConkey *et al.*, 2000; Vassao *et al.*, 2006). Vegetative tissue, in general, releases small quantities of VOCs, which could be induced by mechanical damage and herbivore or pathogen infection (Arimura *et al.*, 2004). In all plant organs, VOC production and emission are developmentally regulated and exhibit similar developmental profiles, increasing during the early stages of development (when leaves are young and not fully expanded, fruits are not yet mature, or before flowers are ready for pollination) and either decreasing or remaining relatively constant thereafter (Dudareva *et al.*, 2000; Vassao *et al.*, 2006). The concurrent temporal changes in activities of enzymes responsible for the final steps of VOC formation, enzyme protein content, and the expression of corresponding structural genes suggest that the developmental biosynthesis of volatiles is regulated largely at the level of gene expression (Dudareva *et al.*, 2000; McConkey *et al.*, 2000; Muhlemann *et al.*, 2012). However, recent data showed that transcriptional regulation of the volatile biosynthetic pathways is not limited to their last biochemical step, but rather shared by one or multiple intermediate steps (Muhlemann *et al.*, 2012).

The common property of VOC blends released from a given plant tissue under different physiological conditions is the presence

of compounds produced by more than one biochemical pathway, formation of which requires coordinated regulation. Orchestrated complex transcriptional regulation is also observed for genes involved in herbivore-induced volatile emissions (Kant *et al.*, 2004; Ralph *et al.*, 2006). To date, only few transcription factors (TFs) regulating the expression of multiple genes in a single biochemical pathway have been identified. Major discoveries have been made in petunia flowers, which are rich in phenylpropanoid/benzenoid compounds. ODORANT1 (ODO1), a R2R3-type MYB TF, is exclusively expressed in petunia petals and controls a major portion of the shikimate pathway as well as the entry points into both the Phe (i.e. chorismate mutase) and phenylpropanoid (i.e. PAL) pathways, thus regulating the precursor availability for the volatile phenylpropanoid/benzenoid compounds (Verdonk *et al.*, 2005). In addition to biosynthetic genes, ODO1 activates the promoter of an ABC transporter of unknown function localized at the plasma membrane (Van Moerkercke *et al.*, 2012). In petunia flowers, ODO1 is positively regulated by EMISSION OF BENZENOID II (EOBII), another R2R3-type MYB TF, which has partially overlapping targets with ODO1 and also activates the promoter of the biosynthetic gene isoeugenol synthase (Spitzer-Rimon *et al.*, 2010; Van Moerkercke *et al.*, 2011, 2012). Unlike ODO1 and EOBII, the MYB4 TF was found to be a repressor of a single enzyme in the phenylpropanoid pathway, cinnamate-4-hydroxylase, and to control carbon flux toward the *p*-coumaric acid-derived volatiles, such as isoeugenol and eugenol, in petunia flowers (Colquhoun *et al.*, 2011).

In contrast to the phenylpropanoids/benzenoids, even less is known about transcriptional regulation of terpenoid biosynthesis in plants. MYC2, a basic helix–loop–helix TF, was shown to activate expression of sesquiterpene synthase genes *TPS21* and *TPS11* in *Arabidopsis* inflorescence by integrating both gibberellic acid (GA) and JA signals into transcriptional regulation of volatile sesquiterpene production (Hong *et al.*, 2012). MYB14 is the first MYB TF directly linked to the control of isoprenoid pathways. It was proposed that MYB14 contributes to the regulation of the volatile terpenoid biosynthesis preferentially via the MVA pathway, as well as JA metabolism during plant defense responses in conifer trees (Bedon *et al.*, 2010). Despite the identification of several TFs for individual pathways, master regulators, which orchestrate formation of diverse volatile blends and act upstream of multiple metabolic pathways, are yet to be discovered.

Transcriptional regulation plays an important role in controlling VOC biosynthesis, but it is not the only mechanism involved. While little is known about post-transcriptional regulation of VOC formation, recent comprehensive reviews provide a thorough discussion of the regulation of the shikimate pathway (Maeda & Dudareva, 2012), as well as the MVA and MEP pathways (Hemmerlin *et al.*, 2012).

The rate of biosynthesis of any particular VOC is not only limited by the activity of enzymes responsible for the final step of its formation, but is rather controlled by the amount of available substrate, especially for enzymes with broad substrate specificities (e.g. some carboxyl methyltransferases and acyltransferases, carotenoid cleavage dioxygenase 1, short-chain dehydrogenase/

reductases) (Effmert *et al.*, 2005; Guterma *et al.*, 2006; Vogel *et al.*, 2008). Precursor availability was also shown to play an important role in the regulation of rhythmic emission of VOCs (Kolossova *et al.*, 2001; Colquhoun *et al.*, 2010; Maeda *et al.*, 2010), as plants often release volatiles with distinct diurnal or nocturnal patterns (Lerdau & Gray, 2003; Martin *et al.*, 2003; van Doorn & Woltering, 2008). However, the actual molecular mechanisms responsible for rhythmic emission of VOCs, frequently controlled by circadian clock (Kolossova *et al.*, 2001; Lu *et al.*, 2002), remain unknown.

Environmental factors such as light intensity, atmospheric CO₂ concentration, temperature, relative humidity, and nutrient status can greatly influence VOC emission (Staudt & Bertin, 1998; Gouinguene & Turlings, 2002). Additionally, successful pollination/fertilization triggers a decrease in overall scent emission mediated by the phytohormone ethylene (Negre *et al.*, 2003; Underwood *et al.*, 2005) and ethylene down-regulates scent biosynthetic gene expression before floral senescence as well (Colquhoun *et al.*, 2010). Signal transduction cascades involved in defense-inducible emission are currently under extensive investigation (summarized in Arimura *et al.*, 2005; Dudareva *et al.*, 2006; Zebelo & Maffei, 2012; Zebelo *et al.*, 2012); however, signaling mechanisms regulating fluctuations in VOCs in response to environmental or physiological factors still await further investigation.

III. Functions of plant volatile organic compounds

Plant VOCs define the chemical landscape of numerous ecosystems wherein they mediate intra- and interspecific interactions. They are notably involved in the attraction of pollinators (Raguso, 2008) and seed dispersers, above- and below-ground defense against herbivores (Degenhardt *et al.*, 2009a; Unsicker *et al.*, 2009; Ali *et al.*, 2012; Hiltbold & Turlings, 2012), protection against pathogens (Huang *et al.*, 2012), and plant–plant signaling (Baldwin *et al.*, 2006). In addition, they protect plants against abiotic stresses such as high light, temperature or oxidative stress (Dudareva *et al.*, 2006; Vickers *et al.*, 2009). The defensive roles of vegetative VOCs are exhaustively discussed in most contemporary reviews (e.g. Unsicker *et al.*, 2009; Mumm & Dicke, 2010; Hare, 2011), and thus here we will focus mainly on functions of floral volatiles in natural ecosystems and their application in agriculture, which are vastly understudied in comparison to their vegetative counterparts. It should be noted that the contribution of VOCs to plant defense and reproduction is species-specific and, in the case of floral volatiles, depends on plant mating system and self-compatibility.

1. Involvement of floral VOCs in plant reproduction

Of all plant organs, flowers generally emit the highest amounts and most diverse blends of VOCs (Knudsen *et al.*, 2006), which function as olfactory cues for the attraction of insect (Dobson, 1994) and bat (von Helversen *et al.*, 2000) pollinators to ensure plant reproductive success. Until recently, the involvement of VOCs in ornithophily remained unclear, as many bird-pollinated

flowers emit negligible amounts of VOCs (Knudsen *et al.*, 2004) and birds likely possess a reduced sense of smell (Bang & Cobb, 1968). However, the exposure of hummingbirds to different volatiles using artificial nectars or transgenic tobacco flowers revealed behaviors reflecting deterrent or attracting effects of scent compounds (Kessler & Baldwin, 2007; Kessler *et al.*, 2008). VOCs act both as long- and short-distance attractants (Dobson, 1994) and provide pollinators with fine-scale spatial information about landing and food opportunities. This fine-scale spatial information relies on tissue-specific and developmentally regulated scent production and emission within the flowers (Dotterl & Jurgens, 2005; Kessler & Baldwin, 2007).

Floral volatile profiles are species-specific and tend to reflect the type of pollinator (Dobson, 2006). For example, plant species pollinated by moths emit high amounts of benzenoids and, to a lesser extent, terpenoids and nitrogen-containing compounds (Dobson, 2006), while bat-pollinated flowers predominantly release sulfur-containing volatiles (von Helversen *et al.*, 2000). Moreover, the emission of VOCs often correlates with the foraging activity of associated pollinators and displays diurnal or nocturnal patterns (Kolossova *et al.*, 2001). Pollinators, on the other hand, are capable of discriminating between complex multicomponent volatile profiles with unique ratios and intensities of compounds (Wright *et al.*, 2005) and therefore develop preferences towards specific profiles (Ashman, 2000). These preferences modulate foraging behavior and promote flower constancy (i.e. the tendency to repeatedly visit flowers with a specific volatile profile), thereby increasing pollen transfer to conspecific plants and preventing stigma clogging with heterospecific pollen (Gruter & Ratnieks, 2011).

Volatile organic compounds emitted from flowers have roles beyond pollinator attraction. Many flower VOCs have antimicrobial or antifungal activities (Hammer *et al.*, 2003; Huang *et al.*, 2012), or function as deterrents against florivores, thus protecting valuable reproductive plant organs (Junker *et al.*, 2011). Moreover, VOCs emitted from pollinated flowers can also repel pollinators and direct them to as yet unpollinated counterparts (Schiestl & Ayasse, 2001). Interestingly, the same VOCs that lure pollinators sometimes serve as signals for floral antagonists. *Silene latifolia* flower VOCs, for example, attract the nursery pollinator *Hadena bicruris*, the larvae of which feeds on and destroys approximately one-quarter of developing seeds, causing a detrimental effect on plant fitness (Wolfe, 2002; Dotterl *et al.*, 2006). In this plant species, a drastic reduction in emission of the key volatiles advertising host location occurs after pollination to minimize further parasitism (Dotterl *et al.*, 2005; Muhlemann *et al.*, 2006). In contrast to *S. latifolia*, where pollinators also act as antagonists, the pollinators and florivores visiting *Cirsium arvense* flowers belong to different insect species (Theis, 2006). In the case of *C. arvense*, diel emission of VOCs correlates with activity of pollinators and is low when florivores are active (Theis *et al.*, 2007). Petunia flowers face the same defense/apparency dilemma as *S. latifolia* and *C. arvense*. To resolve the function of single volatile compounds in deterrence and attraction, five transgenic *Petunia hybrida* RNAi lines silencing individual scent biosynthetic genes were used in field and glasshouse experiments (Kessler *et al.*, 2012).

Rather than reducing scent emission to avoid florivory, petunia flowers specifically emit deterrent compounds such as isoeugenol and benzyl benzoate. Thus, the scent emission profile in the described species is likely defined by opposite selective pressures arising from the necessity to maximize pollinator attraction while avoiding damage by florivores (Muhlemann *et al.*, 2006; Theis *et al.*, 2007; Schiestl *et al.*, 2011). Moreover, recent studies have demonstrated that leaf herbivory alters floral VOC emissions, usually with negative consequences for pollinator attraction (Kessler *et al.*, 2011; Pareja *et al.*, 2012). In wild tomato, this effect is jasmonate-mediated, and therefore floral VOCs are mechanistically tied to phytohormonal pathways, triggering resistance responses (Kessler *et al.*, 2011). This introduces additional challenges for attraction of pollinators while maintaining intrinsic plant defense traits.

Volatile organic compounds released from fruits and seed/spores attract various animal dispersers and are essential for the spatial dynamics of plant populations within ecosystems. In contrast to floral VOCs, a direct involvement of chemical cues in seed–disperser interaction has only been shown in a few instances. A blend of VOCs emitted by seeds of the epiphyte *Peperomia macrostachyam* attracts ant-garden ants *Camponotus femoratus* in the Amazonian rainforest and triggers their seed-collecting behavior (Youngsteadt *et al.*, 2008). By storing seeds in ant gardens where they later germinate, ant-garden ants contribute to seed dispersal. Similarly, fruit-emitted VOCs allow bats to locate ripe fruits (Hodgkison *et al.*, 2007). In the bat-dispersed fig species *Ficus hispida*, volatile profiles are subjected to substantial reconfiguration between the pre-pollination and seed dispersal phases, further supporting the role of VOCs in mammalian-mediated seed dispersal interactions (Borges *et al.*, 2008). Even in nonvascular plants, VOCs can contribute to spore dispersal. In several entomophilic dung mosses, which disseminate their spores by flies, sporophytes produce species-specific odors that attract flies as dispersal agents (Marino *et al.*, 2009).

2. Floral VOCs as drivers of plant biodiversity

The emergence of insect pollination has been proposed to have driven angiosperm diversification in the early Cretaceous (Friis *et al.*, 2006). While the exact mechanisms that led to speciation in early angiosperms remain elusive, evidence from recently evolved species suggests that plant speciation is predominantly the result of prezygotic isolation mechanisms, which prevent mating or fertilization between species (Rieseberg & Willis, 2007). Prezygotic barriers are manifold and include divergent habitat preferences, pollen competition, and pollinator-mediated isolation, the latter being promoted by flower morphology or pollinator behavior. Pollinator behavior, in turn, is predetermined by the pollinator's attraction to specific flower cues. Recent reports suggest that floral VOCs, in conjunction with other flower signals such as color, size and shape, are involved in the evolution and maintenance of reproductive isolation between plant species (Fig. 6). The properties of volatile blends, including intensity, ratios of constituents, and the presence of

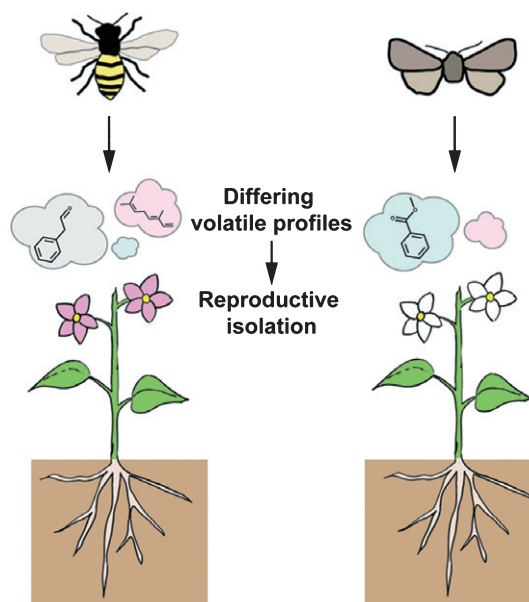


Fig. 6 Role of floral volatile organic compounds (VOCs) in reproductive isolation. In closely related plant species lacking postzygotic mechanisms for the maintenance of reproductive isolation, species integrity is retained by pre-mating isolation. Floral VOCs are an essential component in this isolation. Species-specific floral VOC profiles (illustrated here as differently sized VOC clouds) drive assortative visitation by pollinators, thus preventing gene flow between plant species.

unique compounds, could contribute to the reproductive isolation of closely related plant species (Fig. 6). Sexually deceptive orchids represent a well-studied example of the role of floral VOCs in plant speciation. In this highly specialized system, flowers release VOCs mimicking the female pollinator's sex pheromone, thereby deceitfully attracting male pollinators (Schiestl, 2005). Comparative analysis of floral VOCs emitted by different sexually deceptive orchids of the genera *Chiloglottis* and *Ophrys* revealed that volatiles play a key role in pollinator-mediated isolation. By emitting species-specific volatile profiles, each orchid species in these genera provides a unique attraction channel for a single pollinator species (reviewed in Ayasse *et al.*, 2011). Differences in even a single VOC can strongly influence reproductive isolation between two plant species, as was shown for phenylacetaldehyde in the closely related *Silene dioica* and *S. latifolia*, which emit blends with similar constituents but in different relative amounts (Waelti *et al.*, 2008).

The genetic basis underlying phenotypic VOC differences and ethological isolation between closely related species was recently investigated in different plant systems. Using *Petunia exserta* and *Petunia axillaris*, it was shown that a quantitative trait locus, identified as the MYB transcription factor ODO1, contributes to differences in the scent profile and reproductive isolation between these species (Klahre *et al.*, 2011). In orchids, reproductive isolation among sexually deceptive species of the genus *Ophrys* was proposed to rely on species-specific alkene emission profiles that are defined by differences in enzyme activity and gene expression of a few stearyl-acyl carrier protein desaturases (Schluter *et al.*, 2011; Xu *et al.*, 2012).

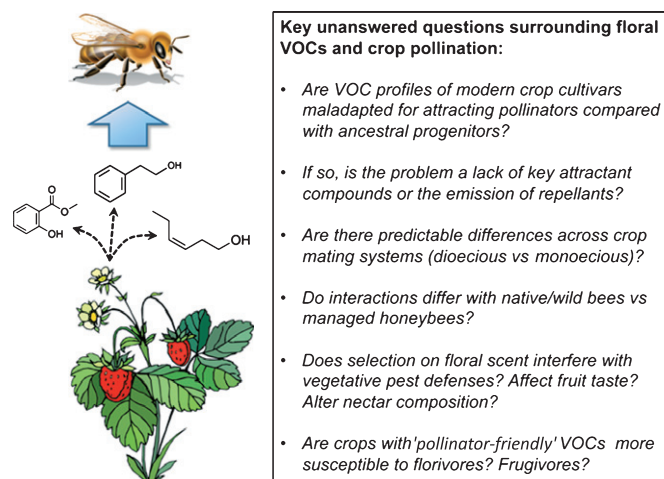


Fig. 7 A list of central questions that should be addressed in future research efforts to enhance crop pollination by manipulating composition of floral volatile organic compounds (VOCs). (Clipart images used with permission from Microsoft.)

3. Impact of floral VOCs on agriculture

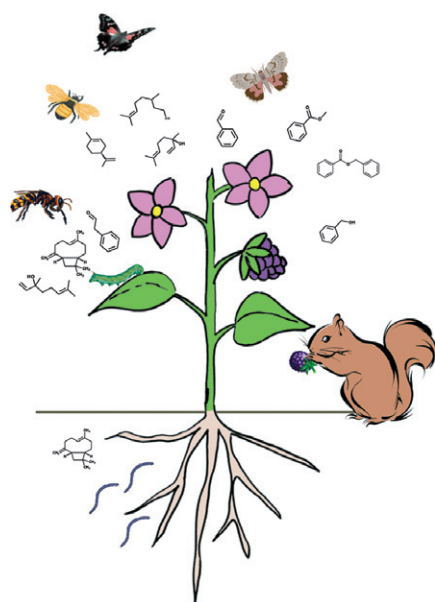
Unlike predators and parasitic wasps whose functional roles could be substituted by chemical insecticides, pollinators cannot be readily replaced. While most crops (e.g. apple, blueberry, canola, cucurbits) benefit to varying degrees from insect pollinators (Losey & Vaughan, 2006), for dioecious crops with self-incompatible flowers it is vital. Thus, floral VOCs have been inadvertently selected to retain some baseline number of pollinator-attracting traits, without which the crop would simply fail to yield fruit. Indeed, comparative analysis of volatiles across several citrus species revealed that self-incompatible pummelo (*Citrus grandis*) blossoms release higher amounts of total VOCs than species (e.g. orange, grapefruit, lemon) with flowers containing both male and female organs (Jabalpurwala *et al.*, 2009). This provides correlative evidence that extreme pollinator dependency has been a driving force to maintain floral VOC profiles in certain crops.

Only a few studies have experimentally evaluated the impact of floral odor on crop pollination (Fig. 7). In one of these studies, bee visitation (number of visits per unit time) was compared between wild-type and *Bacillus thuringiensis* (*Bt*)-expressing eggplant (Arpaia *et al.*, 2011). Although the *Bt* genetic transformation (*cry3Bb* gene) was conducted purely for insect pest resistance, an unintended side-effect was the elevated emission of five (methyl salicylate, Z-jasmone, α -pinene, α -methyl styrene, δ -2-carene) out of 13 total floral compounds from transgenic flowers, all of which elicited positive electroantennogram responses from bee antennae. When both plant types were subjected to commercial bumblebees in a glasshouse, bees strongly preferred *Bt*-eggplant despite significantly smaller flowers and overall plant size, illustrating the dominance of floral scent in pollinator behavior. This study also uncovered an important issue that floral odors likely exist at suboptimal amounts for pollination in many crop species. In addition, it raises the important agricultural question as to whether these suboptimal amounts cause inefficient pollination and subsequent yield loss in commercial fields (Fig. 7).

To date, there are only two examples that demonstrate the link between floral VOC composition and crop pollination failure. Alfalfa seed production, for example, relies on pollination by honeybees, which routinely display low fidelity to alfalfa flowers and, instead, prefer other crops and weedy plants. Low visitation rates and inefficient pollen transfer by honeybees translate into low seed yield. Analysis of behavioral and electrophysiological responses to alfalfa floral VOC chemistry revealed that honeybees, while ignoring major scent compounds (e.g. ocimene), are highly responsive to minor constituents (e.g. linalool) that comprise < 1% of the volatile blend (Henning & Teuber, 1992). As alfalfa flowers also emit deterrents, a generation of new cultivars with elevated emission of attractants and simultaneously suppressed emission of repellants will enhance honeybee visitations to alfalfa. Another example includes glasshouse tomato production, where imported bumblebees are routinely used as pollinators, but preferentially forage on other flowering plants outside the glasshouse. This behavior was observed even when the tomato plants were in full bloom, resulting in poor fruit set and subsequent yield loss (Morse *et al.*, 2012). Glasshouse tomato flowers primarily emit four monoterpenes (β -phellandrene, α -pinene, p-cymene, (+)-2-carene), all of which are known as herbivore-induced leaf volatiles used in defense and possess toxic properties. Emission of these compounds was negatively correlated with bumblebee visitation, suggesting that this largely repellant-dominated floral VOC blend caused pollination failure in glasshouse tomatoes. Taken together, these examples further support the notion that numerous pollinator-dependent crops likely emit suboptimal VOCs (Fig. 7); however, to our knowledge, no breeding programs exist to select for floral fragrances that are 'pollinator-friendly'.

IV. Metabolic engineering of plant volatiles

In natural ecosystems, pollinators often select for elevated floral VOCs (Kessler *et al.*, 2008; Majetic *et al.*, 2009; Parachnowitsch *et al.*, 2012), suggesting that pollination services can be improved by producing more fragrant flowers (Fig. 8). Moreover, many modern crop cultivars are deficient in VOC production as a result of breeding for increased growth or yield which negatively affects secondary metabolism, including VOCs. Indeed, flowers produced by horticultural cultivars of cacao (*Theobroma cacao*) are small, barely fragrant, and only attract opportunistic flies that are inefficient pollinators, whereas wild *Theobroma* sp. possess large aromatic flowers that are predominantly bee-pollinated (Young & Severson, 1994). Similar trends were observed for vegetative VOCs (Fig. 8). Corn domestication resulted in varieties exhibiting a susceptibility to insect pests as a result of the lack of key odorants, primarily (*E*)- β -caryophyllene, that attract pest-killing parasitic wasps (Tamiru *et al.*, 2011) and entomopathogenic nematodes (Degenhardt *et al.*, 2009a). Even more recently domesticated crops display analogous growth–defense trade-offs. Cranberry, for example, is only one or two crosses away from its wild progenitor, *Vaccinium* sp. Yet high-yielding cranberry genotypes have suppressed herbivore-induced sesquiterpene emissions relative to lower-yielding ancestors (Rodriguez-Saona *et al.*, 2011). Taken together, these examples show that crop VOCs are poorly adapted



Advances in metabolic engineering of plant VOCs

Pollination

- Transgenic tobacco with diminished levels of benzyl acetone attract fewer hawkmoths and hummingbirds
- The scent bouquet in rose flowers overexpressing the *Arabidopsis* PAP1 transcription factor is easily distinguished by honeybees

Fruit flavor

- Several attempts have been made to improve flavor of tomato fruits but the effect on fruit/seed dispersers has not been evaluated

Defense

- Patchoulol and additional sesquiterpene products introduced into transgenic tobacco, enhance defense against herbivores
- Linalool production in *Arabidopsis* and potato mitigate direct or indirect defenses
- (3S)-(E)-nerolidol and (E)-DMNT emission from *Arabidopsis* attracts predator mites
- Rice and oregano (E)- β -caryophyllene synthases overexpressed in rice and maize improve plant defense above and below ground

Fig. 8 Advances in metabolic engineering of volatile organic compounds (VOCs). The recent progress in metabolic engineering for improved pollination, plant defense and fruit flavor is documented in a few examples shown in the panel on the right-hand side. Our review also highlights the current lack of knowledge about the influence of VOC alteration on fruit flavor and pollination for improvement of seed (fruit) disperser/pollinator attraction (see Section IV on metabolic engineering).

to maintain beneficial insects in agricultural systems. Additionally, present-day agriculture, while expanding in both size and intensity, tends increasingly to rely on crop monocultures. This reduction in plant diversity translates into adverse effects on ecosystem services provided by the arthropod community, including beneficial insects controlling pests and pollinating crops (Gardiner *et al.*, 2010). Thus, approaches have to be developed to overcome these problems, and metabolic engineering of both floral and defense-related VOCs is a promising tool for enrichment of plant chemodiversity as well as biodiversity of beneficial insect.

In the past decade, numerous attempts have been made to modulate plant volatile profiles and to investigate the effect of changes in volatile emission on insect behavior (Fig. 8). Several strategies were used, including the introduction of new gene(s) or branchways in the host plant, and the modification of existing pathways via up-/down-regulation of biochemical step(s) or by blocking the competing pathways. Indeed, direct plant defense was improved by producing the volatile patchoulol along with additional sesquiterpene products in transgenic tobacco, overexpressing *Pogostemon cablin* patchoulol synthase (Wu *et al.*, 2006). Introduced volatiles deterred tobacco hornworms, a majority of which had migrated away from transgenic to wild-type leaves, consuming 20–50% more of the control leaf material (Wu *et al.*, 2006). In *Arabidopsis*, high concentrations of linalool, achieved by the overexpression of strawberry *linalool/nerolidol synthase* gene (*FaNES1*) targeted to chloroplasts, repelled the aphid *Myzus persicae* in dual-choice assays (Aharoni *et al.*, 2003). Interestingly, elevated concentrations of linalool obtained in a similar way in potato attracted predatory mites and affected tritrophic interactions (Lucker *et al.*, 2006). However, attraction of predatory mites in *Arabidopsis* was attained by producing (3S)-(E)-nerolidol and (E)-DMNT via ectopic expression of the same *FaNES1* gene in mitochondria, which contains the sesquiterpene precursor FPP

(Kappers *et al.*, 2005). Furthermore, when behavioral responses of aphids, parasitoids and predators were compared in three *Arabidopsis thaliana* accessions constitutively expressing *FaNES1* targeted to the mitochondria, different effects were observed depending on ecotypes (Kos *et al.*, 2012). Ecotype-specific function of volatile compounds was also recently recorded for phenylacetaldehyde, which contributed to defense in the Columbia (Col-0) ecotype and was involved in pollinator attraction in Sei-0 and Di-G (Gutensohn *et al.*, 2011). Different effects on plant defense were also observed when rice and oregano (E)- β -caryophyllene synthases were constitutively overexpressed in rice and maize, respectively (Cheng *et al.*, 2007; Degenhardt *et al.*, 2009a; Xiao *et al.*, 2012). In rice, (E)- β -caryophyllene emission improved above-ground plant defense by attracting parasitoid wasps to transgenic plants. By contrast, in a maize line normally lacking (E)- β -caryophyllene and susceptible to western corn rootworm, restored emission of (E)- β -caryophyllene enhanced below-ground defense by providing a signal to entomopathogenic nematodes that infect and kill the root pest (Degenhardt *et al.*, 2009a). These results show that natural defense of crop plants could be improved via metabolic engineering of VOCs, thereby providing an effective and sustainable alternative to current pest-management strategies. However, owing to the species-specific nature of volatile-based plant defenses, results obtained in model systems cannot be directly translated to crops. Moreover, the effect of modified VOC profiles on plant defense needs to be evaluated in an agricultural setting.

In contrast to metabolic engineering of defense-related volatiles, where the effect of altered VOC emissions on insect behavior was investigated, examination of the impact of modified floral scent on animal attraction is in its infancy (Fig. 7). Floral scent bouquet has been modified in several plant species, including petunia (via down-regulation of scent biosynthetic genes (Underwood *et al.*, 2005;

Kaminaga *et al.*, 2006; Orlova *et al.*, 2006), up-regulation of Myb transcription factor PAP1 (Ben Zvi *et al.*, 2008), heterologous expression of scent gene (Guterman *et al.*, 2006) or redirection of carbon flux toward target compound (Koeduka *et al.*, 2008)), carnations (via heterologous expression of scent gene (Lavy *et al.*, 2002) or diversion of carbon flux toward scent formation (Zuker *et al.*, 2002)), and tobacco (via introduction of a novel branchway consisting of three genes (Lucker *et al.*, 2004) or an increase in substrate level (Orlova *et al.*, 2009)). However, the first investigation of the consequences of metabolic engineering of floral VOC on pollinator visitation was performed in transgenic tobacco *Nicotiana attenuata* plants with diminished concentrations of benzyl acetone (Kessler *et al.*, 2008). Monitoring the activity of floral visitors in their native habitat revealed that plants lacking benzyl acetone attracted fewer hawkmoths and hummingbirds than flowers emitting this volatile. It was also shown that changes in more than one volatile compound can influence pollinator behavior. Indeed, remodeling of the scent bouquet in rose flowers overexpressing the Arabidopsis *PAP1* transcription factor was easily distinguished by honeybees (Zvi *et al.*, 2012), the native pollinators of some wild rose species (Shalit *et al.*, 2004). Attractive and repellent functions of different volatiles within the scent bouquet were recently analyzed in petunia flowers using five transgenic lines silencing biosynthetic genes responsible for individual scent compounds (Kessler *et al.*, 2012). While the most highly abundant volatile methylbenzoate acted as both attractant and deterrent, isoeugenol and benzylbenzoate only exhibited the latter function.

Several attempts have been made to improve the flavor of tomato fruits (Fig. 8) (Lewinsohn *et al.*, 2001; Davidovich-Rikanati *et al.*, 2007, 2008). While humans were able to assess changes in fruit aroma, the effect of these genetic manipulations on fruit dispersers has not been investigated.

Overall, the studies described have demonstrated the potential of metabolic engineering of VOCs for both the improvement of plant defense and pollinator attraction (Fig. 8). Despite multiple successes, these results also revealed unexpected problems that can be encountered in VOC engineering: negative effects on plant growth and development as a result of limited carbon availability for essential metabolites or toxicity of the newly introduced compounds (Aharoni *et al.*, 2003, 2006; Orlova *et al.*, 2009); modification of the volatile compounds into nonvolatile forms, for example, by glycosylation (Lucker *et al.*, 2001, 2006; Aharoni *et al.*, 2003); masking by other volatiles (Togni *et al.*, 2010); low to no yield of the desired compound resulting from an insufficient amount of precursors for its biosynthesis (Guterman *et al.*, 2006); and formation of unpredicted compounds as a result of occult metabolic capacities of the host plant (Lewinsohn & Gijzen, 2009). Further, ecological complications will arise once these methodological constraints are lifted. In the wild gourd, *Cucurbita pepo* var. *texana*, for example, experimentally elevated floral emission of 1,4-dimethoxybenzene, considered the most attractive compound to specialist squash bee pollinators, decreased plant reproduction as a result of concomitant attraction of florivorous beetles (Theis & Adler, 2012). Given that *C. pepo* also contains cultivated squash and pumpkin, crops that rely heavily on managed pollinators, genetic enhancement of floral

scent in cucurbits and related species for increased yield may prove difficult. A comprehensive understanding of plant metabolic networks and their regulation, as well as knowledge about key compounds involved in plant–insect interactions in the near future, will greatly advance metabolic engineering of VOCs and help to overcome these challenges.

V. Conclusions

Current advances in chemical ecology highlight the importance of plant VOCs in natural ecosystems as crucial signaling molecules in plant defense, pollination and plant–plant communication. In the past decade, plant volatile research has witnessed a shift from studying VOC composition and the role of individual synthetic compounds in insect behavior toward the elucidation of the intricate interactions between plant-produced volatiles and the native arthropod communities in realistic field settings. Recent breakthroughs in scent gene discovery and metabolic engineering make it possible to manipulate the amount of specific volatiles *in planta*, either omitting or increasing their emission, thus greatly facilitating the identification of key attractants for pollinators and natural enemies. In the future, this transgenic approach will also allow us to determine species-specific functions of different volatiles under varying natural conditions.

Unfortunately, our limited current understanding of the regulation of VOC biosynthetic pathways constrains the manipulation of VOC emission for enhanced pollination and sustainable pest management strategies based on biological control. The emergence of new approaches, however, such as kinetic modeling in conjunction with -omics technologies, will allow us to generate transgenic plants with more precise fine-tuned VOC emissions which will be synchronized with the activities of potential pollinators or herbivores. The identification of transcription factors that regulate the orchestrated emission of volatiles originating from different metabolic pathways will significantly improve our success. Nevertheless, future attempts to increase crop yield will necessitate collaborative efforts by ecologists, entomologists, agronomists, biochemists and plant metabolic engineers.

Another central factor limiting the transition of crops with modified volatile profiles from research laboratories to real-world implementation is the likelihood of nontarget effects. These effects may be unavoidable because most VOCs have multiple functions in plants. A good example is (*E*)- β -caryophyllene, whose function varies dramatically depending on environmental context; from serving as an antimicrobial agent in flowers to reduce bacterial growth (Huang *et al.*, 2012) to acting as an inducible root volatile that attracts nematodes involved in tritrophic below-ground defense (Degenhardt *et al.*, 2009a). To circumvent this barrier, novel creative solutions have to be employed to limit the unintended side-effects of VOCs. Application of -omics approaches, such as transcriptomics and metabolomics, to both plants and insects will undoubtedly promote our understanding of the mechanisms underlying plant–insect interactions and help to achieve desired effects such as maximizing pollination while minimizing pest attraction. Enhancing crop–pollinator interactions becomes even more critical for modern agriculture, which is

facing sudden declines in honeybee populations and is thus increasingly dependent on native bees. Proper selection of emission timing rather than emission quantity and mimicking natural plant 'behaviors' that have presumably been optimized via natural selection over many generations are crucial for obtaining beneficial effects of insects on plant fitness. The successful application of these approaches in agriculture will require additional investigations of basic ecological and evolutionary mechanisms of VOC-mediated interactions in natural plant systems.

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